

Plotting Revenge: What Data tells us about Homeric Retribution

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Abstract

This paper aims to use word embedding models, word frequency, and correlation features to evaluate the ethical landscape of *The Odyssey*, focusing on justice and retribution. *The Odyssey* is an ancient Greek epic poem typically attributed to the poet Homer but is the result of centuries of oral storytelling tradition. While there has been extensive study on retribution in *The Odyssey* as well as how ideas of fate and destiny are represented in Homeric epic, the intersection of these two major themes has been understudied in relation to one another, particularly through a computational lens. This study introduces literary text mining methods as a means to formulate literary arguments grounded in quantitative evidence. Our results showed a strong correlation between ideas of fate and retribution, suggesting that vengeance in *The Odyssey* is carefully planned and morally permissible rather than an emotional reaction to wrongdoing. We hope our results will encourage future projects that use computational methods to provide new perspectives on literary questions in Homeric literature.

1 Introduction

The Odyssey is an ancient epic poem passed down as oral tradition and textualized around the fifth century BCE. The poem is commonly attributed to “Homer” and details the story of a king, Odysseus, returning home from the Trojan War. This paper investigates the question: How can we use word-embedding models, word frequency, and correlation features to understand the role of justice and retribution in *The Odyssey* to evaluate its ethical landscape? Revenge in *The Odyssey* is often narrative, by which we mean that the retaliatory action that an avenger performs— especially killing— that at the same time implies an entire sequence of events that makes the act of killing into “retribution” rather than “murder” (Loney 3). In this paper, we use a range of text mining methods that aim to observe and record the patterns of fate and retribution in the text. This work is essential for reproducibility across the field of literary analysis and produces novel tangible data that augments the existing work of researchers in Homeric studies.

There has been considerable work on the roles of both justice and retribution in *The Odyssey*, generally finding both to be pivotal themes of the epic. Retribution has also been said to be the fulfillment of the cardinal moral rule in Homeric epic: to help one's friends and hurt one's enemies (Stanton, 1984). Some studies also highlight the important role divinity and the gods play in the moral landscape of the epic. Existing literature points to the conclusion that fate and the will of Zeus are identical in Homeric epic (Duffy, 1947). Justice in *The Odyssey*, therefore, is also said to be closely aligned with the gods' will. Retribution from Zeus, which the god himself introduces in the opening book of the poem, offers a moral interpretation of Odysseus' own action: he succeeds because he identifies his purposes with the gods' ways of justice and vengeance (Segal, 1992).

We approach this investigation by using three distinct computational methods, word frequencies, correlation features, and word embeddings. Word frequencies count the number of times a word occurs plotted against narrative time, which is also defined as total running word count. Correlation features are a measure of how variables are related to each other in a dataset and are represented by a value from -1 to 1. Third, word embedding models are a method to represent the semantic meaning of words computationally using vectors in vector space. The vectors permit us to observe the relationship between words with cosine similarity. We found that patterns in the text indicate high correlations between a predestined fate and requital in the text on both a mortal and divine scale, suggesting that vengeance in *The Odyssey* is carefully planned and morally permissible rather than acts of emotion.

2 Data

2.1 Background: Corpus

The Odyssey is an ancient Greek work of epic poetry typically attributed to the poet “Homer.” In reality, *The Odyssey* was the resulting work of an oral cultural tradition that has its roots in the Bronze Age and was likely textualized in the fifth century BCE. In this paper, we will be referencing an English translation of the original ancient Greek work by Robert Fagles (1996). *The Odyssey* is considered one of the foremost classical texts for study due to its range in plot, characters, and its unfading relevance to the modern age (Christensen, 2020). The data we are using for computational analysis is downloaded from the Perseus Digital Library and is the digital corpus of the English translation by A.T. Murray (1919). Although we are using two different translations for this project, all computational data is derived from only the Murray translation.

The Odyssey tells the story of Odysseus who attempts to return home after a ten-year war in the city of Troy. During this journey, Odysseus encounters many obstacles such as monsters, wrathful Gods, and a mutinous crew. At the same time, Odysseus’s son Telemachus and his wife, Penelope, are besieged with raucous suitors who aim for Penelope’s hand in marriage. Apart from homecoming, the poem examines themes of mortality, truth, heroism, and sacrifice. Odysseus represents wisdom and courage, while Penelope is portrayed as the ideal of patience and endurance as she waits for her long-gone husband.

2.2 Data Provenance

All data used in this project was obtained from the Perseus Digital Library (2025). The Perseus Digital Library is an open source offering materials free of charge and their *Odyssey* text is based on public domain editions. Tufts University holds the overall copyright to the Perseus Digital Library; the materials therein (including all texts, translations, images, descriptions, drawings, etc.) are provided for the personal use of students, scholars, and the public. Any commercial use or publication without authorization is strictly prohibited (Perseus Digital Library, 2025).

As a part of the data cleaning process, we are removing all punctuation. We are also lemmatizing the text, meaning reducing each word to its canonical dictionary form in order to reduce variants in the vocabulary. Additionally, we remove stop words, which are very common words that may overwhelm results such as word frequency when performing computational analysis. Finally, we are using pre-processing to remove punctuation, separate sentences, and remove capitalization. These steps are taken to ensure the greatest accuracy and consistency with computational methods.

2.3 Dataset Characteristics

Before any data cleaning steps were taken, our text contained 132646 total words and 6508 unique words. After stop word removal, the text contained 70199 total words and 6385 unique words. Finally, after lemmatization, the corpus contained 70199 total words and 5506 unique words.

Step	Total Words	Unique Words
Before stop word removal	132646	6508
After stop word removal	70199	6385
After stop word removal and lemmatization	70199	5506

Figure 1: Number of Total and Unique words in different stages of data cleaning

3 Methods

3.1 Vocabularies & Distinctive Words

In this paper, we used a variety of computational methods including counting word frequency, training a word embedding model, and using correlation features. To build our corpus, we first download a corpus of the text using the Perseus Digital Library (2025). We then applied several preprocessing steps to clean the text, including lemmatization, stop word removal, and tokenization. These steps were essential for improving the quality of our data by reducing noise and standardizing vocabulary across the corpus.

Word frequency is a method which counts the occurrence of a specific word in a corpus over narrative time, which we define as the cumulative word count across the text. We chose to calculate the word frequency of specific words in *The Odyssey* sectioned by book, meaning our results showed the number of occurrences for specific words in the book of the epic. Word frequency is a useful tool for identifying where specific themes are most prominent, as words associated with a given theme tend to appear more frequently in relevant passages. However, this method is not as useful for data in which the corpus has an incredibly diverse set of vocabulary because word frequencies might be very low and not exhibit obvious trends. There is also the risk of stylistic choice and genre in the text, both of which can heavily affect word frequency and produce misleading results.

In order to compute the word frequency of different words across the books of *The Odyssey*, we used the Python package pandas to store the content of each book in *The Odyssey*. We wrote a function that takes a dataframe and a target word as input and iterates through each book, counting the number of times the target word appears. Our goal was to find which words appeared in specific books and how the frequency of these words varied from book to book. We chose to calculate word frequency for “requital,” “fate,” and “evil” to track the movement of moral ideas related to vengeance over the length of *The Odyssey* and used these words as key markers for these themes. We found that “fate” and “evil” had a higher frequency than “requital” overall, but all three words showed peaks in similar sections of the epic.

Book #	Requital	Fate	Evil	Book #	Requital	Fate	Evil
1	1	1	4	13	1	2	4
2	3	6	11	14	0	2	8
3	1	5	6	15	0	3	5
4	0	6	3	16	0	2	5
5	0	8	4	17	2	4	19
6	0	2	3	18	0	1	8
7	0	2	2	19	0	3	5
8	0	3	2	20	0	1	13
9	0	3	8	21	0	0	3

10	0	3	9	22	2	6	5
11	0	7	6	23	0	2	9
12	0	1	7	24	1	5	7

Figure 2: Table of word frequencies for requital, fate, and evil by book

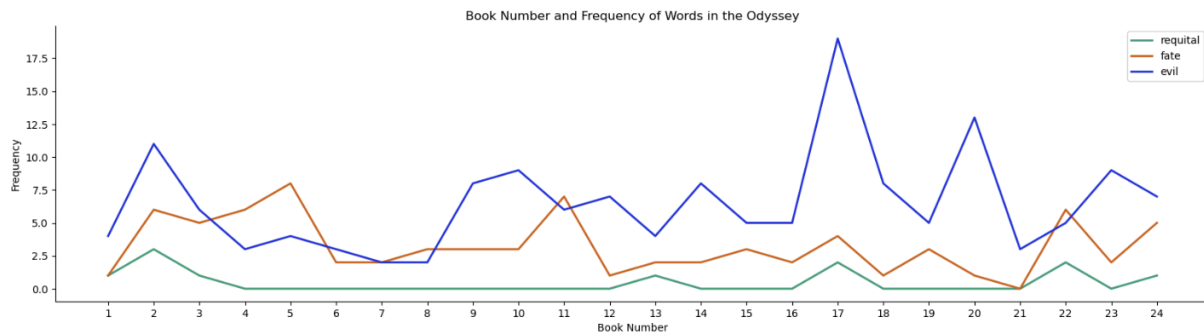


Figure 2.1: Line graph of word frequencies by book for requital, fate, and evil

Words Frequencies in the Odyssey

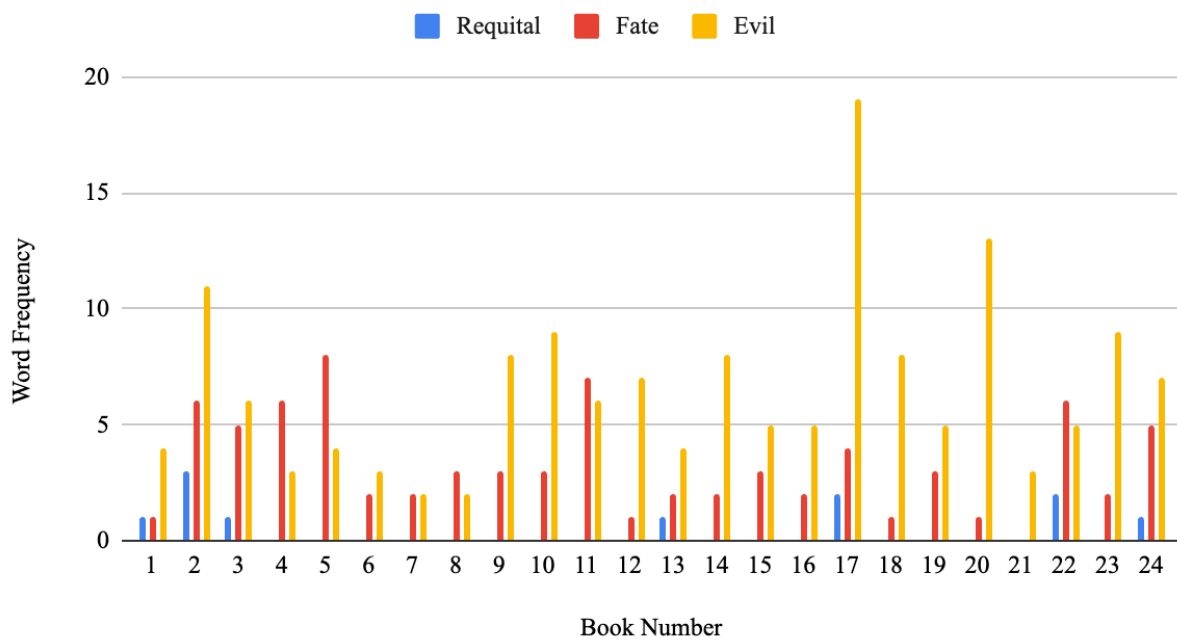


Figure 2.2: Bar graph of word frequencies by book for requital, fate, and evil

3.2 Correlation Features

Correlation features are a measure of how variables are related to each other in a dataset. Correlation features are often measured with a chi squared statistic, which measures the linear correlation between two sets of data. The value of a correlation coefficient ranges from -1 to 1 where 0 is no correlation, 1 is a strong positive correlation, and -1 is a strong negative correlation. A strong positive correlation means that as one variable increases, the other also increases on average. A strong negative correlation means that as one variable increases, the other decreases on average.

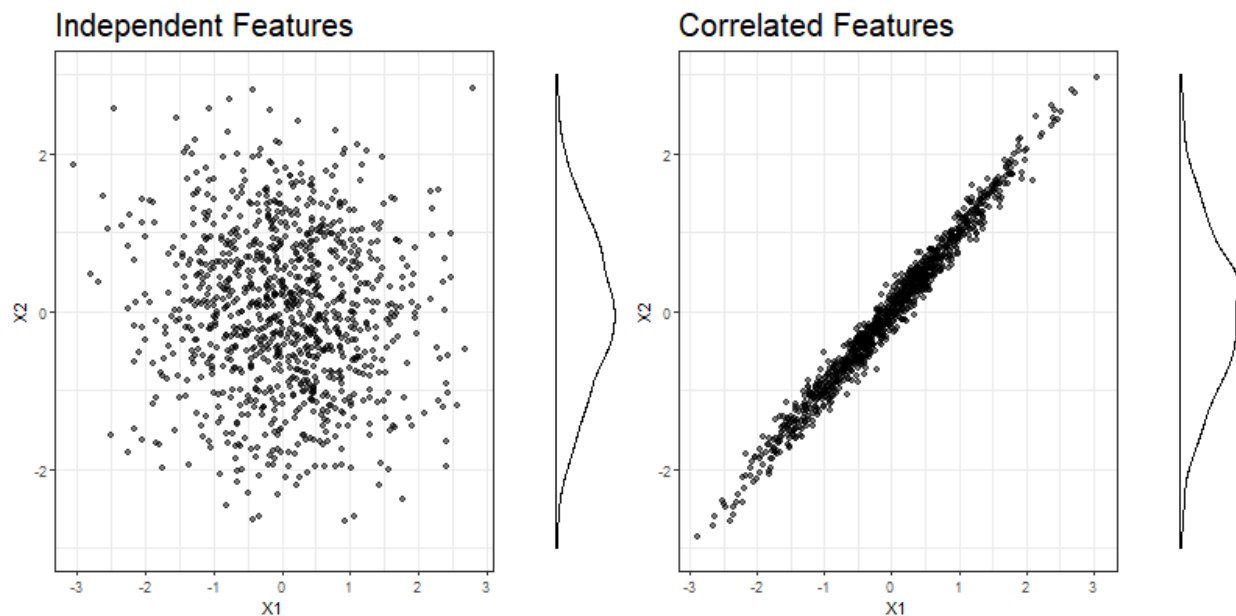


Figure 3.1: Simulated data for independent (left) and strongly correlated (right) features. Source: Kronsender, 2019

Correlation analysis can be a useful method for examining the relationships between words in a text, particularly when the variables being observed exist on a continuous scale. In literary text mining, correlation features encounter problems with vocabulary diversity in a corpus. When the vocabulary of a corpus is extremely large, it is less likely for words to form strong correlations or produce statistically significant results.

We chose to use correlation features to track the relationship between key words in *The Odyssey*. Words that exhibit strong positive or negative correlations suggest thematic associations based on their co-occurrence patterns. In order to produce statistically significant results, we filtered the corpus to only retain the words of relevance and removed all other vocabulary. Without this filtering, all correlations resulted in 0 due to the variance in vocabulary in *The Odyssey*. The correlations were calculated using phi correlation coefficients for a target of five words: "fate", "zeus", "vengeance", "god", "athena," and the Python package yellowbrick. We then displayed the results in a word correlation plot. Due to the sparsity of the corpus after preprocessing, we considered a correlation coefficient of approximately ± 0.5 to ± 0.6 as strong relative to the dataset. We found that "god" and "zeus" had the strongest correlation with a value of -0.49 with "god" and "athena" coming at a close second with a correlation coefficient of -0.36. Another

medium negative correlation were the words “fate” and “god” along with “zeus” and “athena,” both of which had correlations around -0.25. Notably, no correlations were positive.

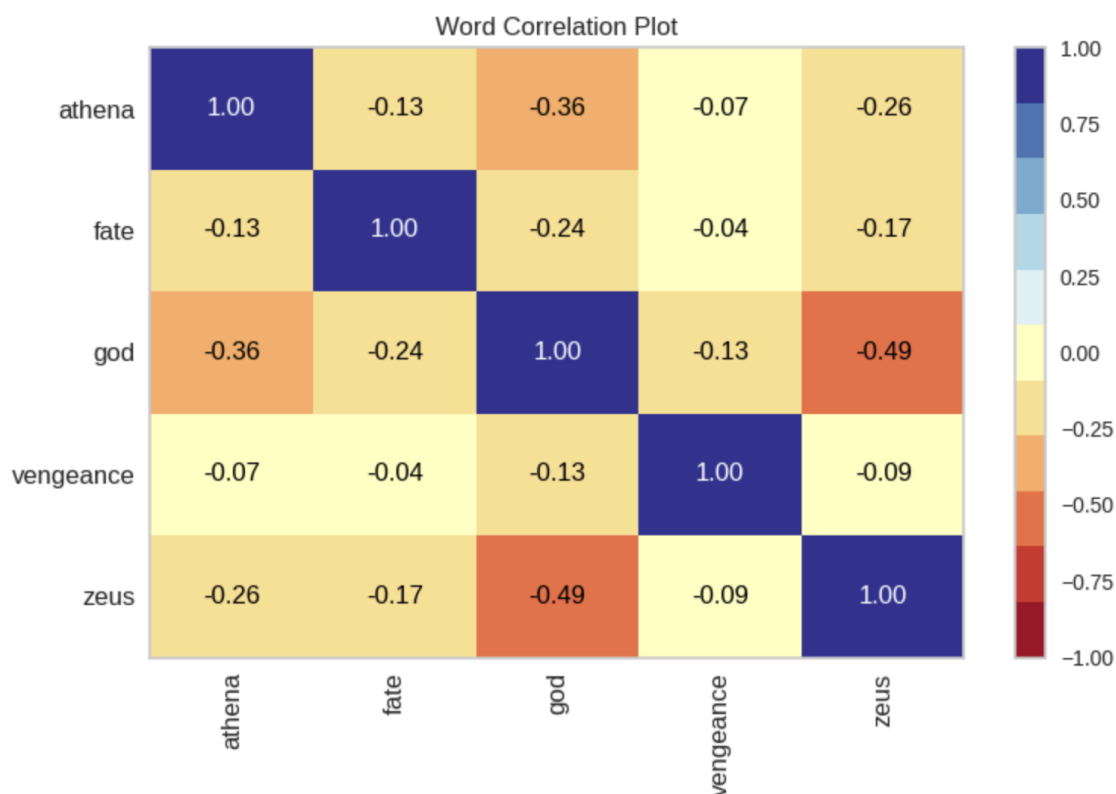


Figure 3.2: Heatmap of word correlations for ‘athena,’ ‘fate,’ ‘god,’ ‘vengeance,’ and ‘zeus’

3.3 Word Embeddings

Word embeddings are a method to represent the meaning of a word computationally. Word embeddings work through a process of vectorization using a word-embedding algorithm. Machine learning models first randomly initialize each word in the corpus and are then trained using a black-box algorithm. This process produces a trained model that can be evaluated using validation questions to assess their accuracy and coherence. Word-embedding models are able to capture and represent the semantic relationships between words, which is useful for executing large-scale semantic analysis of a text and uncovering hidden patterns. The limitations of this method are that it requires a large amount of data to be effective and would not be effective on a subsection of books from *The Odyssey*, for example. Another flaw is that word-embedding models are probabilistic, randomly-seeded models that will produce slightly different results each time and are subject to a certain degree of randomness. For this project, we chose to use the Word2Vec algorithm on our corpus, *The Odyssey*, and elected to use a skipgram model because it tends to perform better for smaller datasets.

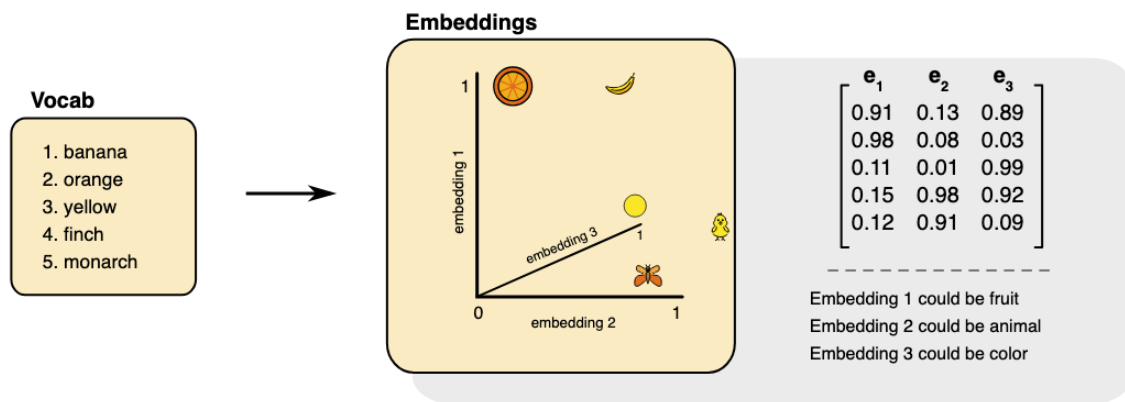


Figure 4.1: Word embedding process for five words. Source: Kharazi, 2024

We chose to train our model over 15 epochs because loss stopped significantly increasing upwards of 15 epochs. We also elected to set the minimum frequency of words to be included in the model at 3 in order to make sure only words of relevance were included and that the word count was low enough to prevent the loss of vocabulary since *The Odyssey* is a small corpus. In order to assess the validity of our model, we used validation questions such as “What are the 10 most similar and dissimilar words to ‘requital’ in *The Odyssey*?” and analogic comparisons. Our results show that the model is well trained as the similar words are closely semantically related to requital.

10 most similar words to “requital”	Distance	10 least similar words to “requital”	Distance
avenger	0.847621440887451	tear	0.010196406394243
gracious	0.847224652767181	shedding	-0.0085179014131
loudthundering	0.840046942234039	flow	-0.036463290452957
violent	0.837318599224090	blood	-0.037363536655902
friendship	0.829792499542236	cave	-0.042245715856552
insolently	0.829610705375671	flung	-0.045218028128147
subdue	0.829404294490814	ran	-0.063414521515369
grant	0.822627246379852	big	-0.064973615109920
wooing	0.818364739418029	around	-0.069321781396865

happiness	0.808963894844055	tall	-0.069498606026172
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Figure 4.2: Top 10 most similar and dissimilar words for ‘requital’

We also conducted three analogies, two for assessing the validity of the model, and a third to gain information about the relationship about vengeance with respect to morality in *The Odyssey*. The first analogy created was athena : zeus :: odysseus : ?, to which the model provided “prince” as an answer. This is highly consistent with the expected answer, as “prince” likely refers to Telemachus who is Odysseus’s son, just as Athena is Zeus’s daughter. The second analogy we used was ship : sea :: pear : ?, to which the model predicted “fig.” This also confirms the validity of the model, since a fig is something found near to a pear, just as a ship is something found near the sea. With our similarity testing and analogies, our validation questions were successfully answered and confirmed that the model could be used for further literary analysis. We ran an analogy to determine the relationship between vengeance and the moral framework of the epic by posing the equation honor : right :: vengeance : ?. The model provided ‘pitiless’ as an answer.

Using our trained word embedding model, we evaluated the cosine similarity of pairs of words and performed vector math. To analyze the cosine similarity, we used the vectors created by the model and calculated the cosine of the angle between those two vectors. A cosine similarity of -1 indicates that the words are extremely dissimilar while a cosine similarity of 1 indicates that the words are highly similar. The cosine similarity for “justice” and “vengeance” was 0.5976 and the cosine similarity of “hero” and “vengeance” was 0.6120. Vector math is another form of analogy reasoning which uses word embeddings against specified target words to determine the best answer, similar to a multiple choice quiz. This was calculated by taking the vectors of given query words and performing addition and subtraction operations to result in a new answer vector. This answer vector was used to calculate the cosine and euclidean distance between the answer vector and each vector associated with the target words. The closest target word is returned as the answer. When given the equation fate + justice - pity, the model produced “requital” as the answer from the words ["requital", "honor", "rest", "peace"]. We also displayed our embedding results using the projector Tensorflow software. The total variance of the projection was about 38.6% with dimensions conformed to 100.

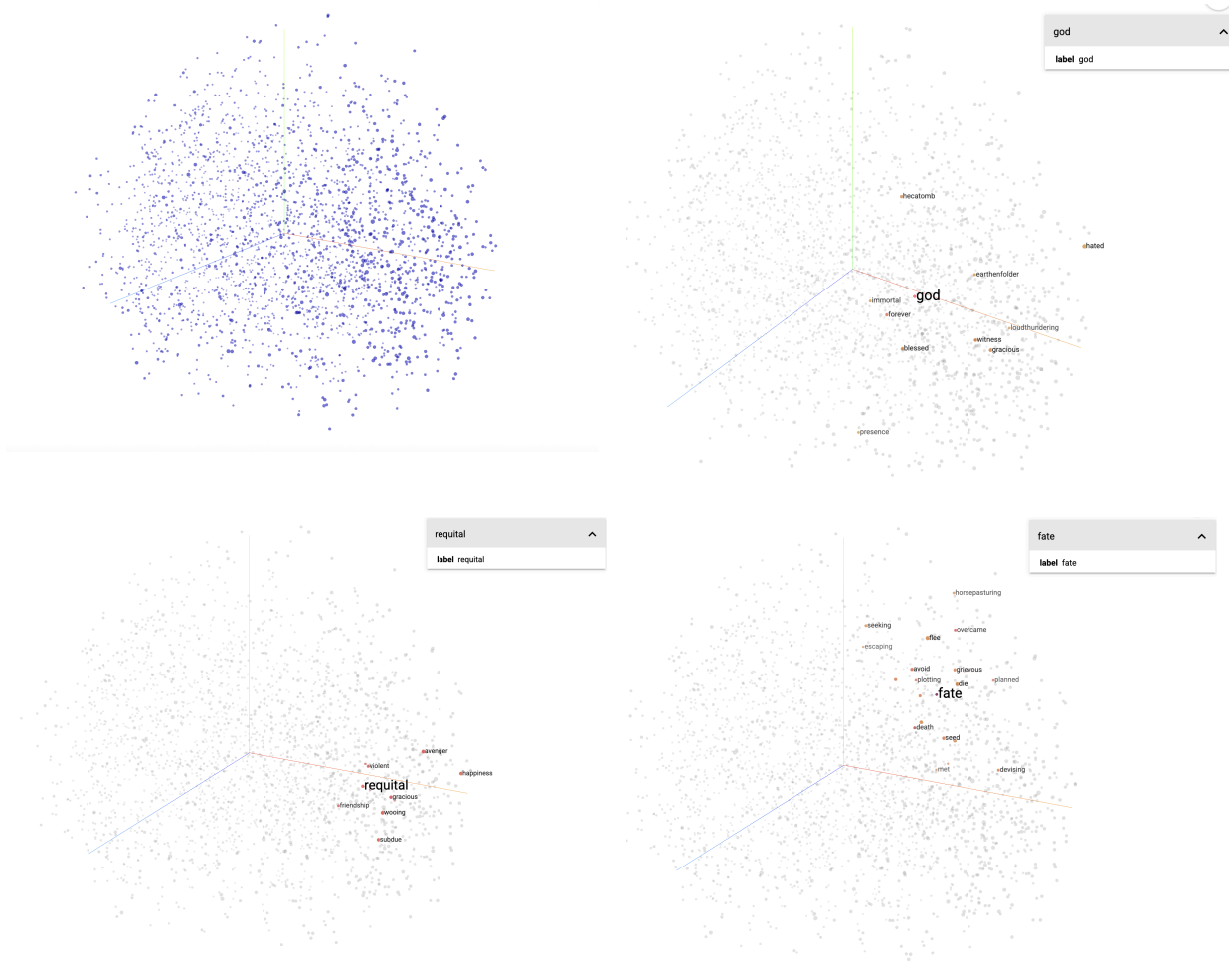


Figure 4.3: Word embeddings for *The Odyssey* corpus (top left), 10 closest words to ‘god’ (top right), 10 closest words to ‘requital’ (bottom left), 10 closest words to ‘fate’ (bottom right)

4 Literary Analysis

Revenge and retribution are central to *The Odyssey*, driving many of the integral plot points and themes of the book. Although modern ideas of revenge are often accompanied by images of vigilante superheroes and a level of self-interest (Kort-Butler 2012), vengeance is widely used and accepted in Homeric society. In fact, ideas of justice were based on the idea of reciprocity, and reciprocity meant revenge (Stanton 1984). Another prominent recurrence in *The Odyssey* is the presence of fate. Scholars have strongly debated the role of fate ruling over the gods or the same as the will of the gods, specifically Zeus (Duffy 1947). In either case, fate is an everpresent player in the lives of the characters in *The Odyssey*, and the events of the epic are directly connected to these higher powers. We argue that justice and retribution are seen as integral parts of society in *The Odyssey* precisely due to fate; methods of vengeance are the fulfillment of the predestined consequences that accompany every action. In this paper, we will focus on three prominent scenes of retribution in *The Odyssey*—Odysseus’s slaughtering of the suitors, Prince Orestes’ killing of Aegisthus, and the death of Odysseus’ crew by Zeus—to illustrate the patterns of vengeance that are seen throughout the book. We can use word frequencies, word embedding models, and correlation features to illustrate patterns in the text that show a strong relationship between fate and requital in the text on a mortal and divine scale, suggesting that vengeance in *The Odyssey* is carefully planned and morally permissible rather than an emotional reaction to wrongdoing.

Word frequency over the books of *The Odyssey* indicate that revenge is preordained and predictable rather than arbitrary. The epic foretells the death of the suitors from the introduction of the novel, with Zeus’s discussion of Agamemnon’s death as an example of mortal folly (Homer, 1996). The glorification of Orestes’s avenging of his father plants the seeds for a parallel method of avenging honor in Odysseus’s home. In addition, the punishment for the suitors is clearly described in Book 2 of *The Odyssey*, where the suitors are warned that Odysseus is coming soon and a bloody death is imminent (Homer, 1996). This scene shows that vengeance is not a surprise, but is clearly defined and planned prior to execution. Using calculations of word frequency, we can see that “fate,” “evil,” and “requital” are seen to peak in the same sections of the epic, indicating a relationship between the three events. Building on Segal’s work on divine justice in *The Odyssey*, we can observe that retribution follows a formulaic pattern of warning, transgression, and retribution in order to restore balance to society (Segal 1992). The cattle of the sun are a good example of this formula, as Odysseus is explicitly warned by both Circe and Tiresias to leave the cattle alone, lest his crew be doomed (Homer, 1996). Despite this warning, the crew commits the transgression of eating the cattle despite, importantly, the availability of fish and other food on the island (Segal 1992). This moral culpability leads to their destruction at the hands of Zeus to give adequate punishment for the men’s sins. This pattern of events is reproduced in the guilt of the suitors and Aegisthus’s betrayal of Agamemnon which warrant the ultimate demise of these characters.

Using a trained word embedding model on *The Odyssey*, we determined that acts of vengeance are viewed as morally permissible and expected, rather than vindictive acts of personal emotion. Using the Word2Vec algorithm, we trained a CBOW model on a corpus of *The Odyssey* and then used cosine similarities and vector math to evaluate the ethical landscape of the text. We found that the words “vengeance” and “justice” had a cosine similarity of 0.5976 indicating a strong positive semantic relationship between the two concepts. “Hero” and “vengeance” similarly displayed a cosine similarity of 0.6120. Furthermore, when given the equation fate + justice - pity and the target words "requital", "peace", "rest", "thee," the model produced the answer “requital” using both cosine similarity and Euclidean distance. These results indicate that semantically, requital and a sense of moral correctness are present in *The Odyssey*. The model was also able to complete the analogy honor : right :: requital : ? and produced “pitiless” as the result, suggesting that in the epic’s context, mercy is not praised. Instead, revenge and retribution are prioritized and there is no room for lesser consequences because the balance of the society would be thrown off. Odysseus’s slaughter of the suitors at the end of the epic despite their pleas for mercy supports this conclusion (Homer, 1996). His unrelenting brutality emphasizes the necessity of justice at any costs and his status as the hero of the epic reinforces the moral validity of his actions. This can be also attributed to the emphasis on honor and heroism, which would justify Odysseus’ violent slaughter of the suitors as a necessary step to defend his honor and demonstrate his strength in the face of a challenge, even worth risking his own life (McHardy, 1999). In a similar vein, Orestes’ revenge on Aegisthus is praised by the gods, specifically Zeus and Athena, with Athena using the language, “what glory Prince Orestes won throughout the world when he killed that cunning, murderous Aegisthus” (Homer, 1996). The tone of her speech, emphasizing glory and heroism, are a clear indication that she wants Odysseus to follow in his footsteps and is attempting to entice him to take action. In the aftermath of the cattle of the sun, Helios demands to be paid back “in blood” for the murder of his cattle to which Zeus acquiesces and kills the guilty crew (Homer, 1996). Helios’s immediate demand for blood retribution and the fact that Zeus himself obliges, often thought to be the embodiment of fate’s will, shows that retribution is divinely ordained and therefore the correct mode of action (Duffy 1947).

Finally, we must evaluate the relationship between divinity and the moral framework of *The Odyssey*, which we elected to do using a word correlation plot. In order to ensure that our data was usable, we created a separate corpus of *The Odyssey* only preserving the words used in the correlation plot to account for vocabulary diversity. Then, using the words "fate", "zeus", "vengeance", "god", "athena," we created a heatmap of word correlations that displayed strong relationships between the gods, fate, and vengeance. The strong negative correlation between Zeus & Athena compared to “god” can be explained by the fact that the two words are likely interchangeable and therefore will not occur in the same contexts in the epic. Zeus has a correlation of -0.17 and Athena has a correlation of -0.13 with the word “fate” while the word god itself has a correlation of -0.24 with “fate.” This again indicates a link between these variables, which can be accounted for by portraying the gods as the ones who create fate or by

the gods themselves participating in the system of retribution and therefore executing fate in the epic. The latter is supported in Odysseus' speech to the suitors where he accuses them of having "No fear of the gods who rule the skies up there, no fear that men's revenge might arrive someday" (Homer, 1996). Here, Odysseus is the actor of justice but he clearly points to the gods as the origin of that justice. The killing of the suitors, undertaken with the sanction of Zeus and the assistance of Athena, has a character more of execution than of voluntary slaughter, supporting the idea that the suitors' own actions are the cause for their doom (Stanton 1984). Similarly, Orestes's revenge and Zeus's revenge for the cattle of Helios are directly sanctioned by the gods, making them the determiners of morality in the Homeric world. Man must curb his desires recognizing that "one can not, howsoever valiant he be, thwart the purpose of Zeus who is mightier far" (Keith 1928). The gods are seen to also demarcate the boundaries of retribution in *The Odyssey*, such as when Athena orders Odysseus to seek peace at the end of the epic rather than perpetuate the cycle of revenge started by the suitors.

Using computational evidence, we were able to track and evaluate the moral framework of justice and retribution in *The Odyssey* and determined that retribution is closely intertwined with fate. This work establishes the difference in perceptions of vengeance in Homeric society compared to the modern world, with Homeric society embracing vengeance as a method to restore balance to the world. It also proves that ideas of revenge in *The Odyssey*—and perhaps in the ancient Greek world—are more complex than the common saying "an eye for an eye," suggesting a carefully wrought social system which has unspoken repercussions for every transgression. Our computational methodology lays the framework for further investigation into the moral framework in *The Odyssey*, potentially in the original Greek text. Despite the results being statistically weaker than expected, we must take into account the diversity in vocabulary and variation in the text that leads to weaker correlations and lower frequencies of words. In the context of these factors, therefore, we can reframe a strong correlation and a "high frequency" relative to the other words in the text.

5 Conclusion

In this paper we sought to use word embedding models, word frequency, and correlation features to understand the role of justice and retribution in *The Odyssey* and how that forms the ethical landscape of the epic. We calculated the frequency of “requital,” “fate,” and “evil” and found that the three words peak in the same sections of the epic poem. Our analysis of correlation features for five words, “fate”, “zeus”, “vengeance”, “god”, “athena,” showed a high negative correlation between words related to the concept of god and fate. Finally, we used the cosine similarities of vectors in our word embedding models to find the relationship between requital and moral value in the epic. Our analysis showed that requital is viewed as the “correct” solution for transgressions across *The Odyssey*. We used literary evidence from three pivotal scenes of retribution in *The Odyssey*, Orestes’s revenge, Odysseus’ slaughter of the suitors, and the death of Odysseus’s crew after they killed the cattle of the sun god. Our results indicate that requital in *The Odyssey* is pre-ordained and morally permissible due to the strong relationship between the idea of fate and vengeance. This, in addition to the carefully formulated steps for revenge replicated in the various scenes of *The Odyssey*, verifies that revenge is a result of express permission from the Gods and is ethically justified.

6 Ethics & Limitations

There are many ethical considerations that must be taken into account when performing research with literary text mining methods. Text mining can be an effective way to map changes in culture ideas, linguistic evolution, and the topics of interest at a certain point in history. For example, the word “slavery” peaked in frequency during the Civil War era, and then had a resurgence during the civil rights movement around 1955, indicating the political climate of the United States during those time periods. Text mining is also able to track linguistic change: as language evolves, the frequency of words and phrases rises and falls due to cultural phenomena, leading to changes in grammar as British English fades out and slang is incorporated into the lexicon. Text mining can also be incredibly useful in detecting repression and censorship throughout history, a prominent example being Nazi repression during the World War 2 era. In the realm of literature, text mining can be a tool to examine works at a scale that would be impossible for a human being. For example, as Ted Underwood presents in his blog post, the claim that William Wordsworth emblemizes Romantic alienation can be supported by text mining that finds words like “solitary” and “alone” throughout his poems (Underwood, 2015). Computers allow people to find large-scale patterns that cannot be found in ordinary reading- patterns that span a much larger corpus and require more computational power than a human brain could handle. A third and enticing benefit to text mining is the idea of modeling. With a model, scientists can represent a phenomenon, the variables that may cause it, and draw conclusions about its stability over time (Michel et al., 2010). Machine learning techniques have also made it possible to create unsupervised models, where a learning algorithm finds clusters or patterns rather than the reverse. In this method, researchers can discover patterns unnoticed by the human eye and in directions previously untapped.

On the other hand, text mining comes with sizable flaws and limitations. A prominent one is the lack of digital data that can be used for quantitative methods. In the study conducted by Jean-Baptiste Michel et al. (2010), the researchers used a subset of 5 million digitized books, which represents only 4 percent of the total books ever published. Only 12 percent of books have been digitized, meaning a large majority of published books are inaccessible to text mining methods and could lead to skewed or biased results. The resulting corpus used in the study also contains a majority of English, French, and other major languages which can restrict the scope of events studied and remove minority accounts of history. Another limitation of text mining is that the results cannot be taken at face value. When looking at common 1-grams in the English language, text mining produces nonword categories. These could be 1-grams with nonalphabetic characters, misspellings, and foreign language words. When researchers manually annotated samples from the lists of common 1-grams, they found that around 51% of words in 1900 to 31% of words in 2000 were members of the nonword categories, significant numbers that can skew conclusions drawn from the data. Misspellings represent words that should be taken into account when calculating the frequency of words, something that computational methods overlook. A third flaw in text mining methods is the lack of contextual insight data can provide in comparison to human analysis. While text mining can provide useful information on a diction level, these insights may not necessarily translate into insight on tone, theme, and imagery for example. Underwood’s blog post (2015), which gives the example of Romantic poetry and William

Wordsworth, presents this idea that the frequency of words may have more to do with their common use in a language rather than something specific to that author's style.

Another broad concern with text mining is the ethics of data itself. McKittrick (2021) makes the argument that algorithmic models are based in inherently racist, sexist, and colonial discourse. Statistics on the rate of women making false rape arguments falls in line with her idea that these algorithms and statistics ignore the totality of life and experience, reducing experiences to numbers that calculate negative outcomes by default. Particularly, she would argue that these statistics would lead to models that anticipate women to lie and frame women as suspicious despite claims of scientific neutrality. When pointing out the flaws in algorithms with regards to black death, McKittrick writes that "The question of why or how or under what conditions premature death is not asked...Anything in excess of violence, racism, and subordination—black life, black joy, the practice of loving, for example—is either absent or perversely tied to the dehumanizing logics of white supremacy" (2021). This same rationale can be applied to false allegations by women- the conditions or stories behind a case of false reporting is not asked and anything apart from the narrative of a woman telling a lie is absent. The thousands of true cases, where women's lives and stories are valued and believed, are not captured to the algorithm leading to even biological conclusions that cause computers to inherently classify women as liars. Although in this project we do not engage with data concerning real people, these considerations surrounding the inherent biases in datasets must be taken into consideration.

We also engage in the phenomenon of distant reading in this project, a topic which has been highly contested and criticized by literary critics who prefer the traditional form of close reading for literary argument. Our form of analysis most closely aligns with Hannu Salmi (2023), who understands reading at scale as identifying smaller nuances in the text that are amplified when massive amounts of data are collected and analyzed, usually over a much larger time frame. She also highlights that distant reading can "be seen as an array of strategies in organizing data and thus also preparing it for other reading methods, such as close reading" (Salmi, 2023). Salmi (2023) highlights the digital visualization tools and techniques that use technology to assess big data in her definition of distant reading. Ted Underwood (2017), on the other hand, contends that reading at scale is not separate from close reading, but simply integrates a longer timeline incorporating samples from the past to explore a previously uncharted territory. The evidence in this approach is not, as Moretti and other definitions of distant reading suggest, a maximal approach to encompass as much text as possible. Instead, Underwood (2017) emphasizes the importance of sampling over longer time periods and credits computational methods for giving literary scholars new questions and to theorize existing questions in new ways. In distant reading, with Underwood's definition, interpretation happens with a sociological approach to literature involving methods, then results, then conclusions as opposed to thesis-driven literary tradition. In this paper, we did not engage with Underwood's form of distance reading, choosing to construct a thesis and then applying distant reading methods to find evidence to support our argument.

Yet another significant concern lies in the treatment of literature as data. In Stephen Marche's (2012) article, "Literature is not Data: Against Digital Humanities", he argues against turning literature into data. Throughout the essay he notes how the culmination of literary data itself is a

product of multiple intellectual failures. The use of massive online databases, stylometry and distant reading there has been an unprecedented loss of library culture. He states in general that digital humanities are a ‘failing comedy’ in itself and that, “literature cannot meaningfully be treated as data” (Marche, 2012). The act of turning literature into data has a reductionist nature removing the distinction, the taste, the ability to criticize and the historical reception of the works. He rounded this argument out by stating, “Algorithms are inherently fascistic” (Marche, 2012). In conclusion, he argues that algorithms cannot replace human nuance and interpretations. Meaning is meant to be mushy and insight remains handmade.

Our method of word frequency also has limitations on the extent of conclusions that can be drawn from frequency alone. Although frequency is an extremely simple metric with powerful affordances, incorrect conclusions can be drawn if literary analysis is not taken into account. David McClure’s (2017) blog post, “Distributions of words across narrative time in 27,266 novels” is an example of using word frequency calculations to make generalizable insights about the form and structure of novels. However, without the use of literary evidence from human reading, which is impossible for a corpus of that size, misguided conclusions can easily be drawn to justify patterns in the data.

Our other method of analysis, word embeddings, is a more complex technology that also comes with a host of criticisms. For one, word embeddings “pick up the biases contained in this data, which may be of gender, race or culture” (Prakul, 2024). When training the model on a corpus like *The Odyssey*, for example, it can pick up the inherent biases in the text and lead to conclusions based on these biases during literary analysis. Researchers have also pointed out flaws in the ways that word-embedding models are evaluated for validity, specifically word similarity tasks. An example is that the notion of word similarity is subjective and is often confused with relatedness (Faruqui et al., 2015). For example, the word ‘cup’ and ‘coffee’ are often presented as semantically similar, but they refer to two very different kinds of objects. Another criticism is that no strong correlation is found between the performance of word vectors on word similarity and extrinsic evaluation NLP tasks like text classification, parsing, sentiment analysis (Tsvetkov et al., 2015; Schnabel et al., 2015). This means that despite a model passing similarity tests, it may not perform well on the actual task the model is asked to perform. Finally, researchers have found that models tend to ignore polysemes, or words that have multiple meanings in algorithms like skip-gram which were used in this project (Faruqui et al., 2015). This could pose significant problems for our work because *The Odyssey* is a translated corpus and the meaning of many words could be radically altered due to the author choosing to use a specific word in English. This problem may be reduced with work in the original Greek corpus or by implementing new research on multi-dimensional vectors to capture word meaning (Reisinger and Mooney, 2010).

7 Future Work

Literary text mining is not without its challenges and limitations, many of which are apparent when working with a corpus as small as *The Odyssey*. One significant challenge was the discrepancy in translation between the two texts we were working with. We performed all close reading and non-computational literary analysis using Fagles' translation of *The Odyssey*, but all computational work was performed on the online corpus of Murray's *Odyssey* translation. These two works were composed almost 70 years apart and this leads to significant variation of vocabulary. For example, the word "suitors" in the Fagles translation was consistently replaced with "wooers" in Murray's translation. This, along with many other examples, posed a barrier in being able to compare word frequency data between the two translations. Another challenge with this project is the diversity in vocabulary used in *The Odyssey*. This variability led to weak correlation values between words as well as low frequencies that rendered the strength of our results feeble. Due to this limitation, we chose to reframe the boundaries for statistically significant results, looking at the data in context to other parts of *The Odyssey* rather than all literature in general.

This project is only an introductory look into the relationship between fate, retribution, and morality in *The Odyssey*. There are many more computational methods than can be applied to *The Odyssey* to find more patterns such as topic modelling and stylometrics. Additionally, there is significant scope and benefits to repeating this analysis with the original Greek text, which may impact variations in vocabulary and produce stronger results. Another extension of this project could be to observe at what point in narrative time retribution takes place to observe if there is any relationship between the timeline of events in the epic and the theme of revenge. We are still uncertain why all correlations resulted in negative values and would have to conduct a more comprehensive correlation feature analysis to discover any pairs of words that exhibit positive correlations. We also remain uncertain if our word embedding model was trained to produce the highest quality of results. We would need to conduct many more rounds of testing and model evaluation to find model conditions that could produce more accurate results, perhaps with different algorithms such as FastText or Glove.

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